The vector $b=\begin{bmatrix}4\\1\\-17\end{bmatrix}$ in the plane is spanned by $a_1=\begin{bmatrix}1\\4\\-2\end{bmatrix}$ and $a_2=\begin{bmatrix}-2\\-3\\7\end{bmatrix}$. $\square$

18) Let $v_1=\begin{bmatrix}1\\0\\-2\end{bmatrix}$, $v_2=\begin{bmatrix}-3\\1\\8\end{bmatrix}$, and $y=\begin{bmatrix}h\\-5\\-3\end{bmatrix}$. For what values of h is y in the plane generated by v_1 and v_2?

$\boxed{\text{Soln:}}$ The plane generated is actually the span of the vectors v_1 and v_2. By defn. of Span, y is in the plane generated if y is a linear combination of the vectors v_1 and v_2. That is, $x_1v_1+x_2v_2=y$ for some $x_1,x_2\in\mathbb{R}$. This is equivalent to claiming that the augmented matrix $\begin{bmatrix}v_1 & v_2 & y\end{bmatrix}$ has a solution set.

$$\begin{bmatrix}1&-3&h\\0&1&-5\\-2&8&-3\end{bmatrix}\xrightarrow[+2\times \text{Row}1]{\text{Row}3=\text{Row}3}\begin{bmatrix}1&-3&h\\0&1&-5\\0&2&2h-3\end{bmatrix}\xrightarrow[\text{Row}3-2\times\text{Row}2]{\text{Row}3=}\begin{bmatrix}1&-3&h\\0&1&-5\\0&0&2h+7\end{bmatrix}$$

<u>Verification that the final matrix is in echelon form</u>

1. If $2h+7\neq0$, then there are no zero rows in the matrix. $\therefore$ The condition that all non-zero rows must be above all zero rows is vacuously true.

If $2h+7=0$, then the third row is a zero row. The first two rows are non-zero rows. $\therefore$ The condition that all non-zero rows must be above all zero rows is True.

2. Row 1 Leading entry $=1$ in Column 1
Row 2 Leading entry $=1$ in Column 2
Row 3 possible Leading entry $=2h+7$ in Column 3

$\therefore$ The condition that the leading entry of each row must be in a column to the right of the columns of the leading entries of all rows above it is True.